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BEHAVIOURAL CHARACTERISTICS AND INVOLVEMENT IN DIFFERENT TYPES OF TRAFFIC ACCIDENT

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Abstract—Recent research has shown risk of road traffic accident involvement to be associated with the tendency to commit driving violations, fast driving, and a lack of thoroughness in decision making. It has also been shown that three broad types of accident can be used to categorise more than 70% of drivers' brief written descriptions of their accidents. These categories are "shunts" (rear-end collisions), right-of-way violations, and loss-of-control accidents. Furthermore, for each accident type the role of the reporting driver may be defined as active (striking) or passive (struck) depending on the position of his/her vehicle. This paper reports two studies examining the extent to which measures of tendency to commit violations (measured by the Driver Behaviour Questionnaire (DBQ)), frequency of fast driving (measured by the Driving Style Questionnaire (DSQ)) and thoroughness in decision making (measured by the Decision Making Questionnaire (DMQ)) are associated specifically with involvement in each of the main accident types. The results showed that a high DBQ-violation score was associated with accidents in general, both active and passive, and specifically with active loss-of-control accidents and passive right-of-way accidents. High DSQ-speed and low DMQ-thoroughness were associated with active but not passive accidents, high DSQ-speed with active right-of-way violations, and low DMQ-thoroughness was specifically associated with active shunts and active right-of-way violations. The implications of these results are discussed.

Keywords—Traffic accident, Driver behaviour, Behavioural characteristics.

INTRODUCTION

Recent research has helped to provide a clearer understanding of the association between behavioural and psychological characteristics of drivers and their risk of traffic accidents (see Elander, West, and French 1992; Grayson 1991; Grayson and Lester 1990 for comprehensive reviews of the substantial body of research that has addressed this topic). Such associations may be helpful in designing global accident countermeasures where more focused action in terms of enforcement or changes to vehicle and road characteristics is not possible. For example, it may be difficult to legislate adequately to prevent drivers from driving too fast for the current traffic conditions but within the posted speed limit or from moving into the path of other vehicles. However, a more global change in driver behaviour fostered through improved training or a different set of attitudes might reduce accidents.

Among other findings, tendency to commit driving violations, fast driving, and lack of thoroughness in decision making have all been reliably shown to be associated with increased accident risk (Reason et al. 1991; West, Elander, and French, 1991). For example, Reason et al. (1991) showed that drivers who reported a relatively high level of violations during the course of their driving tended to be overrepresented among the accident involved, while there was no significant association between self-reported errors while driving and accident involvement. Violations were defined as deliberate deviations from safe driving practice such as running red lights. Errors were defined as driving mistakes or omissions such as forgetting to check the rear view mirror. This finding lent support to one possible view of traffic accident causation, namely that a large proportion of accidents may arise from wilful violations of good driving practice rather than lack of skill or tendency to make mistakes.

The findings of West et al. (1991) were also consistent with this view, focusing specifically on driving speed as an important correlate of accident risk. Both self-reported and observed driving speed were shown to be correlated with accident risk with accidents being recorded both retrospectively and prospectively. Although the association between speed and accident risk was already well established, this study showed that it was independent of miles travelled, driving experience, age, and sex.

West et al. (1991) also found evidence that self-reported lack of thoroughness in decision making, not only in the driving domain, was associated with increased accident risk, partly mediated by faster driving speed but independent of age, sex, driving experience, and miles travelled.

However, it is one thing to show associations between accidents and behavioural characteristics and another to show that the behavioural characteristics contribute to accident risk. Ideally, one could establish that behavioural characteristics contributed to accident risk by manipulating the former in an experiment and observing an effect on the latter. In the case of driving speed, it is well known that interventions to reduce speed in particular locations can reduce accident rates. However, a causal link between a general tendency of drivers to drive fast and a higher accident rate has not been tested in this way. Thus many drivers report that they are "fast but safe." Indeed, faster drivers tend to be more confident and have a higher opinion of their driving skill. They believe that they can handle the speeds at which they drive. The question is whether this is, in fact, the case or whether fast drivers are at increased risk because of the speeds at which they drive. Similarly, although many accidents are obviously caused by blatant violations of the highway code or the law, a tendency to commit violations may be associated with an increased risk of accidents of other types. Given that a large proportion of accidents do not involve deliberate lawbreaking, it would be important to know whether this was the case.

Experimental tests of the possible contribution of behavioural characteristics such as fast driving, tendency to commit violations, or careless decision making to accidents would be extremely expensive and difficult to mount. However, a weaker test of a possible causal link consists of examining whether behavioural characteristics are associated with particular types of accident for which a plausible causal mechanism can be established. For example, if the characteristic of fast driving is a direct cause of increased accident risk, its effect should be detectable specifically on accidents in which excess speed could have been a contributory factor. If it turned out that fast driving was linked with accidents in which no plausible link with speed could be inferred, this would imply that fast driving was but a marker of some other characteristic that increased risk of accident.

Similarly, it is not difficult to imagine how lack of thoroughness in decision making might contribute to accident risk. Failing to look carefully before pulling out into traffic or to consider the consequences of continuing to drive fast in slippery road conditions could lead to many different types of accident. However, previous research has shown that lack of thoroughness is associated with fast driving and that this mediates some but not all of its link with increased accident risk. Lack of thoroughness has also been shown to correlate with social deviance, which is itself linked with higher accident risk (West, Elander, and French 1993). So the role played by lack of thoroughness in elevating the risk of accident is still unclear.

Perhaps the strongest a priori case for a direct link between behavioural characteristics and accident risk arises with driving violations (Parker et al. 1995; Reason et al. 1991). Such behaviour (e.g. running red lights and giving chase to other drivers) is outlawed precisely because it is dangerous. Indeed, there are clearly cases of accidents preceded by outright disregard for driving regulations. There is evidence that drivers who report that they are more disposed to commit violations regard the potentially negative consequences of such violations as less important, perceive less social disapproval for the commission of violations, and feel that such behaviour is less under voluntary control (Parker et al. 1992). Furthermore, these drivers show evidence of a false consensus bias, making estimates of the number of other road users who commit violations that are higher than those made by drivers who do not commit violations (Manstead et al. 1992).

However, many traffic accidents probably do not involve deliberate and blatant violations of the accepted conventions of driving and it is possible that tendency to commit violations is a marker of a more widespread disregard of risks associated with driving and lack of concern for one's own and other's welfare that would put drivers at risk of any kind of accident, whether or not a clear violation was present. Moreover, tendency to commit violations is closely linked with faster driving and it may be that this in itself is the key factor for most drivers (West et al. 1993).

Teasing apart the links in this complex network of associations is likely to prove extremely difficult. However, it may be possible to go some way towards this goal by looking at specific associations between particular behavioural or psychological factors and different kinds of accident. For example, if it were found that faster driving speed was associated only with accidents in which speed could be considered to be a causative factor, it would be reasonable to assume that tendency to drive fast was directly linked with greater accident risk. Conversely, if it were found that tendency to drive fast was associated with accidents in which drivers pulled out when they did not have right of way, this would suggest that it was not speed so much as some more general characteristic of carelessness or impatience that was important. In fact, studies have shown that younger drivers are more likely to be killed in single-vehicle, rollover crashes (Evans 1991), whereas older drivers are at relatively greater risk of injury from multi-vehicle, side-impact collisions (Viano et al. 1990). This would be consistent with the former type of accident being associated with tendency to drive too fast and the latter type being associated with a failure of observation.

Apart from helping to establish links between behavioural characteristics and risk of involvement in different types of accident, understanding those links should also prove useful for development of countermeasures. It is one thing to be able to inform drivers that they run an increased risk of accident, but common sense would suggest that it is much more helpful to know specifically which types of accident various types of driver might be susceptible to, in order to target training, education and enforcement.

A system for classifying accidents based on brief written reports by drivers has recently been devised (West 1993). The system involves assigning the accident to one of several "scripts" based on the physical movement of the vehicle(s) involved, and assigning an "active" versus "passive" role to the reporting driver based on the part played by his or her vehicle. While there is no obvious way in which the drivers' descriptions could be checked by objective recording of the accidents, there are several features of the classification system that suggest that it is reasonably valid. First of all, the categories involved are sufficiently distinct that it would require complete fabrication of accident details by drivers to move an accident from its true category to another (e.g. from a shunt to a right-of-way violation). If drivers were unwilling to provide at least an approximate account of their accidents, one would imagine that they would simply leave the description blank or, more likely, not return the questionnaire. If drivers were tending to bias their accounts in order to minimise their own role in the accident, one would expect to end up with far more "passive" than "active" accidents. (The distinction between passive and active is based primarily on whether the reporting driver would be likely to be held responsible in law.) In fact, this is not what was observed. Third, to the extent that the classification system contains random errors of classification, there would be a failure to observe relationships between demographic characteristics and involvement in specific accident types. In fact, it has been shown that young males are more at risk of certain types of accidents, defined using this classification system than others (West 1993).

Therefore, although there is considerable scope for error and bias in using drivers' brief written reports to determine accident type, the above considerations suggest that with large samples, it should be possible to detect specific patterns of relationship between behavioural characteristics and accidents of different types. Preliminary support for this view comes from a study in which self-reported fast driving speed as a general behavioural characteristic was shown to be specifically associated with what were judged independently to be "speed-related" accidents (West et al. 1991).

The present paper describes two studies. One examined relationships between scores of a sample of drivers on the Driver Behaviour Questionnaire (DBQ; Reason et al. 1991) and involvement in particular kinds of accident as measured on two separate occasions. The DBQ contains three subscales: driving errors, driving violations, and driving lapses. It had previously been shown with this sample that DBQ-violations rather than DBQ-errors were significantly associated with accidents per year taking account for demographic variables and annual mileage (Parker et al. 1995). The first study reported here examined the extent to which this link was focused on particular accident types.

The second study examined the relationship between self-reported fast driving and lack of thoroughness in decision making on the one hand and involvement in particular kinds of accident reported in a separate accident survey, on the other. In the sample used, both behavioural characteristics had been shown to correlate with accidents in general, taking account of annual mileage and demographic characteristics. The purpose of this study was to examine how far the behavioural measures were linked to particular accident types.

STUDY 1: RELATIONSHIPS BETWEEN ERRORS, LAPSES, VIOLATIONS AND ACCIDENT INVOLVEMENT

Method

Relationships between self-reported driving behaviour and rate of involvement in different types of accident were examined. The measure of driving behaviour used was the Driver Behaviour Questionnaire (DBQ), which provided scores on three subscales: errors, violations, and lapses (Reason et al. 1991). These were the main predictor variables. Accidents divided into the five most common types recorded in the 1987 and 1990 surveys acted as outcome variables. The accident types used were: active shunts (rear-end collisions in which the responding driver does the hitting), active right-of-way violations, active loss-of-control accidents, passive shunts (rear-end collisions in which the responding driver is hit), and passive right-of-way violations (see below for definitions). Estimated miles driven over the accident reporting periods, age, sex, and years of driving experience were used as covariates.

Sample

Drivers taking part in a postal national accident survey in 1987 provided brief written descriptions of accidents in which they had been involved. The sample for that survey consisted of four separate subsamples: a random sample of drivers aged over 22 years (random old), a sample of drivers aged over 22 years who had had a full driving license fewer than 10 years (inexperienced old), a random sample of drivers aged over 60 years (elderly) and a sample of drivers aged under 23 years (young). The young drivers sample was selected to contain 50% 17- and 18-year-olds and 50% in the 19–22-year-old group. All the samples contained equal numbers of males and females. The samples were obtained from the U.K. Driver and Vehicle Licensing Agency. The young drivers' questionnaire included questions on age, months of driving experience, sex, and number of accidents since passing the driving test. The questionnaire for the other samples was similar except that it requested years of driving experience and number of accidents in the preceding three years. Accidents were defined as collisions causing damage or injury that occurred while driving a vehicle on a public highway, and so included a high proportion of damage-only accidents.

Full data were obtained from 19,068 drivers (Maycock, Lockwood, and Lester 1991), a response rate estimated at between 75% and 80%. Two years later, a subsample of 2187 drivers from the original 19,000 were sent the DBQ. A total of 1656 drivers responded positively. Of those 1656, 1373 also responded in the final accident survey in 1990. The sample for the present study comprised these respondents. Thus, for each of those 1373 drivers, we had information on their accident record over a total of six years, plus a self-report measure of their driving behaviour, collected separately, during the middle of that six-year period.

Given that not all the respondents who were sent the DBQ responded in the second accident survey, a comparison was made on key variables recorded in the first accident survey and on the DBQ between those who did and did not respond. These are given in Table 1. It is clear that responders to the second accident survey differed in a number of respects from nonresponders: they were older, more likely to be male, and had more years of driving experience, lower DBQ-violations scores, fewer passive and active accidents, and, in particular, fewer active loss-of-control accidents. In general this would work to truncate the distribution of accidents and accident-related variables so that any estimate of relationships between these variables using the sample that responded in both surveys would be conservative.

Table 2 compares accident rates across the two accident surveys for those drivers who responded on both. It is apparent that there was a marked reduction in rates of both active and passive accidents and in particular active loss-of-control accidents. Thus even among the relatively safer group of drivers who responded to both accident surveys, there was a reduction in accident rates over time.

Measures

Full details of the system for classifying accidents in the 1987 and 1990 accident surveys are given elsewhere (West 1993). For the present purposes, the five most common accident types were used. These were active shunts, active right-of-way violations, active loss-of-control accidents, passive shunts, and passive right-of-way violations. These were defined as follows:

Shunt: One vehicle hits another on the same carriageway from behind; active if reporting driver's vehicle does the hitting and passive if reporting driver's vehicle is hit.

Right-of-way violation (ROWV): A vehicle pulls onto or across a carriageway without right-of-way; active if reporting driver's vehicle commits the violation and passive if reporting driver's vehicle is on carriageway with right of way.

Loss-of-control accident: A driver fails to control the direction of a vehicle and keep it on the carriage-

Table 1. Study 1: Characteristics of those who did and did not respond to second accident survey

Variables recorded during first accident survey and behaviour survey	Responders to all surveys[1], $N = 1373$	Those who failed to respond to second accident survey, $N = 283$	Sig. of difference[2]
Mean age (SD)	41.5 (16.03)	38.1 (16.08)	$p < .001$
Mean years' full license (SD)	16.0 (12.49)	13.2 (16.08)	$p < .001$
Proportion of males	50.3	42.4	$p < .05$
Mean annual mileage × 1000 (SD)	8.4 (9.20)	8.8 (10.58)	ns
Mean DBQ error score (SD)	.00 (.99)	.02 (1.05)	ns
Mean DBQ violation score (SD)	−.03 (.98)	.16 (1.06)	$p < .01$
Mean DBQ lapse score (SD)	.01 (1.00)	−.07 (.98)	ns
Mean active accidents (SD)	.21 (.54)	.30 (.63)	$p < .01$
Mean passive accidents (SD)	.19 (.49)	.26 (.56)	$p < .02$
Mean active shunts (SD)	.05 (.24)	.06 (.26)	ns
Mean active ROWV (SD)	.03 (.18)	.04 (.20)	ns
Mean active loss-of-control (SD)	.05 (.25)	.10 (.34)	$p < .01$
Mean passive shunts (SD)	.08 (.31)	.10 (.33)	ns
Mean passive ROWV (SD)	.04 (.22)	.04 (.19)	ns

[1]These formed the subjects for the present study.
[2]By t-test in the case of quantitative variables such as age and chi-squared test in the case of proportions.

way; active if reporting driver's vehicle goes out of control.

These five types accounted for 63% of accidents in the first accident survey. There were not enough accidents of the other types to provide confident estimates of parameters.

The DBQ is described in full elsewhere (Reason et al. 1991). Respondents indicate how often they commit each of a set of 24 actions, examples of which are given in Table 3. A 6-point response scale is used, where 1 = Never, 2 = Hardly ever, 3 = Occasionally, 4 = Quite often, 5 = Frequently, 6 = Nearly all the time. Factor analysis of responses to this instrument across several studies suggests that within that list of behaviours there are three reliable independent factors. The first factor, labelled DBQ-errors includes errors made while driving. The second factor, DBQ-violations comprises deliberate violations of normal safe driving practice. The third factor, DBQ-lapses consists of relatively harmless mistakes resulting in the main from lapses of attention. Rolls, Hall, Ingham, and McDonald (1991) have reported a high level of correspondence between self-reported driving behaviour as measured by the DBQ and observed driving behaviour on a 40 km test route.

The DBQ subscale scores referred to in the following section are factor scores and thus have a mean of 0 and standard deviation of 1. Such factor scores can be taken as a measure of overall tendency to commit violations, make errors, and suffer lapses. Negative scores represent below average values and positive scores represent above average values.

Results

The main analyses were multiple regressions, with number of accidents of particular types as outcome variables. The three DBQ subscale scores, age at time of first accident survey, sex, years with full driving license at time of first accident survey, and

Table 2. Study 1: Mean accident rates (standard deviations in parentheses) in first and second accident surveys among those who responded to both surveys

	First accident survey $N = 1373$	Second accident survey $N = 1373$	Significance of change[1]
Active accidents	.21 (.54)	.13 (.39)	$p < .001$
Passive accidents	.19 (.49)	.15 (.41)	$p < .01$
Active shunts	.05 (.24)	.04 (.21)	ns
Active right-of-way violations	.03 (.18)	.02 (.15)	ns
Active loss-of-control accidents	.05 (.25)	.02 (.16)	$p < .001$
Passive shunts	.08 (.31)	.06 (.27)	ns
Passive right-of-way violations	.04 (.22)	.04 (.21)	ns

[1]By Wilcoxon test.

Table 3. Study 1: Examples of items from the Driver Behaviour Questionnaire

Item	Subscale
Drive especially close to the car in front as a signal to its driver to go faster or get out of the way	Violation
Become impatient with a slow driver in the outer lane and overtake on the inside	Violation
Cross a junction knowing that the traffic lights have already turned against you	Violation
Disregard the speed limits late at night or very early in the morning	Violation
Failure to notice that pedestrians are crossing when turning into a side street from a main road	Error
Fail to check your rear-view mirror before pulling out, changing lanes, etc.	Error
Underestimate the speed of an oncoming vehicle when overtaking	Error
Brake too quickly on a slippery road or steer the wrong way in a skid	Error
Attempt to drive away from the traffic lights in third gear	Lapse
Switch on one thing, e.g. the headlights, when you meant to switch on something else, e.g. the wipers	Lapse
Realise that you have no clear recollection of the road along which you have just been travelling	Lapse
Get into the wrong lane approaching a roundabout or junction	Lapse

estimated mileage over the accident reporting period concerned served as predictor variables and were entered together, to assess the extent to which DBQ scores were associated with accidents over and above demographics.

Mileage was calculated by taking the annual mileage reported in the first survey, multiplying this by the number of years over which accidents were reported in that survey, and then adding annual mileage reported in the second survey multiplied by three. (The second survey always covered a three-year reporting period.) This was necessary because for the young driver sample, the reporting period for the first survey was the number of years since they had passed their test and this was not always the same.

In all multiple regresssion analyses a Poisson distribution was assumed for the accident frequency data. In the tables that follow, associations between predictor variables and accident frequency are expressed as rate ratios. These indicate the change in accident rate per unit change in the predictor variable. A rate ratio of 1 indicates no change; a rate ratio of .5 indicates a halving of the rate; a rate ratio of 2 indicates a doubling of the rate. The form of the model is $\lambda = e^{\mathbf{z}'\beta}$ where λ is the expected accident rate per year, $\mathbf{z}$ is the vector of covariates (values of predictor variables) and β is the vector of corresponding regression weights being estimated and whose antilogs produce the rate ratios. For each estimated regression coefficient a $\mathbf{z}$ statistic is produced by dividing the regression coefficient by its standard error the square of which can be compared with the chi-squared distribution with one degree of freedom.

In the interests of clarity and economy, only those associations that were statistically significant will be reported. Thus Table 4 shows that the DBQ-violations scale was related to accidents in general, active accidents overall, active loss-of-control accidents, passive accidents overall and passive right-of-way violations. There was no statistically significant association between DBQ-violations and active or passive shunts, or active right-of-way violations. The DBQ-error scale was associated only with active accidents in general. The DBQ lapse scale was not associated with any type of accident at all. Thus, DBQ-violations were specifically associated with active loss-of-control accidents and passive right-of-way violations and this relationship remained once age, sex, and years with a full license were taken into account.

The possibility of an interaction between DBQ-violations and age or experience was investigated by computing two new variables representing age multiplied by DBQ-violations and years of experience multiplied by DBQ-violations. These new variables were added one at a time to the multiple regression already including DBQ-violations, age, sex, mileage, and years of experience. Including either

Table 4. Study 1: Prediction of accidents from demographic variables plus Driver Behaviour Questionnaire factor scores

Accident type	Predictor variables	β	SE of β	Rate ratio
All accidents	DBQ-violations	.15	.04	1.18***
All active accidents	DBQ-errors	.14	.05	1.15**
	DBQ-violations	.14	.05	1.15**
Active loss-of-control	DBQ-violations	.20	.10	1.22*
All passive accidents	DBQ-violations	.18	.05	1.20***
Passive right-of-way violations	DBQ-violations	.20	.10	1.22*

Note: ***$p < .001$; **$p < .01$; *$p < .05$.

Table 5. Study 1: Results of multiple regression in which DBQ-violations by age interaction was added to the list of predictor variables

	Active accidents			Active loss of control accidents		
Predictor variable	β	Se of β	Rate ratio	β	Se of β	Rate ratio
DBQ-errors	.14	.05	1.15**	.17	.10	0.90
DBQ-violations	.41	.11	1.25***	.75	.25	2.11**
DBQ-lapses	.08	.05	1.08	−.09	.10	0.92
DBQ-violations* age	−.10	.04	0.99*	−.24	.01	0.98*

Note: *$p < .05$; **$p < .01$; ***$p < .001$.

DBQ-violations by age or DBQ-violations by experience added significant predictive power to the multiple regression in the case of active accidents in general and active loss-of-control accidents in particular.

Table 5 shows the results when the DBQ-violations by age interaction term was added. The results when the DBQ-violations by experience interaction was added were identical. The fact that the rate ratio for the interaction term was less than 1 indicates that the relationship between DBQ-violations and accident risk became less acute with increasing age and experience. However, it was not possible to tell whether age or experience was the critical variable.

STUDY 2: SPEED, THOROUGHNESS IN DECISION MAKING AND ACCIDENT RISK

Method

A subsample of 980 of the drivers who had responded to the 1987 accident survey (described in Study 1) completed questionnaires in 1989, which included scales concerning fast driving and thoroughness in decision making. This subsample (described in detail in West et al. 1991), was stratified according to age, sex, annual mileage, and accident involvement. The aim was to provide a wide range of accident risk for males and females, young and old, and high- and low-mileage drivers, equally. Seven hundred and eleven drivers responded, of whom 693 provided complete data for the present study. Their characteristics are shown in Table 6. None of the drivers approached to take part in Study 2 had been approached for Study 1.

Measures

Fast driving and lack of thoroughness in decision making were assessed using the questions shown in Table 7. Full details of the questionnaires of which they formed a part are given elsewhere (West et al. 1991). The driving speed scale was drawn from the Driving Style Questionnaire (DSQ) and the thoroughness scale was drawn from the Decision Making Questionnaire (DMQ). Accidents were recorded and categorised as described in Study 1. Self-reported driving speed measured by the DSQ has been shown to correlate well with observed driving speed on a test route involving motorway, rural single carriageway, and urban single carriageway roads (West et al. 1991).

Table 6. Study 2: Characteristics of sample

Characteristic	Mean N = 693	SD
Age in 1987	37.2	19.2
Proportion of males (%)	51.7	
Years with full licence in 1987 accident survey	13.5	15.0
Annual mileage 1987	8518	8878
Active accidents	0.31	.60
Passive accidents	0.24	.48
Active shunts	0.09	.31
Active right-of-way violations	0.04	.20
Active loss-of-control accidents	0.08	.30
Passive shunts	0.10	.31
Passive right-of-way violations	0.05	.23
DSQ—Speed (0–15 scale)	6.30	3.91
DMQ—Thoroughness (0–20 scale)	15.00	3.2

Table 7. Study 2: Items used to measure speed and thoroughness

Item	Scale
Do you exceed the speed limit in built-up areas?	DSQ-speed
Do you exceed the 70 mph speed limit during a motorway journey?	DSQ-speed
Do you drive fast?	DSQ-speed
Do you make decisions without considering all the implications?	DMQ-thoroughness
Is your decision making a deliberate logical process?	DMQ-thoroughness
Do you plan well ahead?	DMQ-thoroughness
Do you work out all the pros and cons before making a decision?	DMQ-thoroughness

Table 8. Study 2: Prediction of accidents from demographic variables plus scores on Driving Style Questionnaire and Decision Making Questionnaire

Accident type	Predictor variables	β	SE of β	Rate ratio
All accidents	DSQ-speed	.03	.01	1.03*
	DMQ-thoroughness	−.89	.20	0.41**
All active accidents	DSQ-speed	.05	.02	1.05*
	DMQ-thoroughness	−1.61	.04	0.20***
Active shunts	DMQ-thoroughness	−1.67	.75	0.19*
Active right-of-way violations	DSQ-speed	.11	.05	1.11*
	DMQ-thoroughness	−2.59	1.11	0.07*

Note: $^*p < .05$; $^{**}p < .01$; $^{***}p < .001$.

Results

The main analyses were multiple regressions using number of accidents of particular types as dependent variables. The speed or thoroughness scores, mileage, and age, sex, and years of driving experience were used as predictor variables and entered together to assess the extent to which scores were associated with accidents over and above demographic variables.

Mileage was calculated by taking the annual mileage reported in the accident survey and multiplying this by the numbers of years over which accidents were reported in that survey. This was necessary for the reasons already provided in connection with Study 1.

As in Study 1, a Poisson distribution was assumed for the accident frequency data and associations between predictor variables and accident frequency are expressed as rate ratios to indicate the change in accident rate per unit change in the predictor variable. A rate ratio of 1 indicates no change; a rate ratio of .5 indicates a halving of the rate; a rate ratio of 2 indicates a doubling of the rate.

Nonsignificant associations have again been omitted from the table for the sake of clarity. There was no significant association between DSQ-speed and passive accidents in general, active or passive shunts, active loss-of-control accidents, or passive right-of-way violations. Nor was there a significant association between DMQ-thoroughness and passive accidents in general, active loss-of-control accidents, passive shunts, or passive right-of-way violations. Table 8 shows that speed was associated with accidents in general, active accidents overall, and active right-of-way violations, and that these associations were over and above any contribution of demographic variables. Thoroughness of decision making was independently associated with accidents in general, active accidents overall, active shunts and active right-of-way violations. Neither DSQ-speed nor DMQ-thoroughness showed significant interactions with age or experience in predicting numbers of accidents.

It has been shown previously that DSQ-speed and DMQ-thoroughness are negatively correlated (West 1991). Therefore, it was necessary to assess the independent contributions of these factors to prediction of accident rates. Thus a subsidiary analysis was carried out in which both were entered in a forced entry multiple regression along with mileage, age, sex, and driving experience. DMQ-thoroughness emerged as an independent predictor of active accidents (rate ratio = 0.23, $p < .001$), active shunts (rate ratio = 0.22, $p < .05$) and active right-of-way violations (rate ratio = 0.11, $p < .05$). DSQ-speed was not significantly associated with any specific accident type.

Thus, DMQ-thoroughness was associated with active accidents generally, active shunts and active right-of-way violations even with mileage, experience, age, sex, and DSQ-speed taken into account. However, there was no clear evidence that DSQ-speed was associated with particular types of active accident once demographic variables and DMQ-thoroughness had been taken into account.

DISCUSSION

The results support the view that particular behavioural characteristics would be linked to involvement in some types of accident and not others. However, the picture that emerged was somewhat different from what might have been expected a priori.

DBQ-violations were associated equally with both active and passive accident involvement. There are two possible explanations. First of all, it may be that passive involvement in accidents should not be construed in terms of simple "bad luck," just happening to be in the wrong place at the wrong time, but is in part attributable to an aspect of driving style that is reflected in DBQ-violations. Secondly, it is possible that drivers high on DBQ-violations distorted their accident descriptions so that they were coded as passive when in fact they were active. In support of the former hypothesis is the finding reported elsewhere that individuals involved in what are deemed to be passive accidents of various types in one reporting period are at higher risk of involvement in similar accidents in a second reporting period (West 1993), even once age, sex, driving experience, and annual mileage are taken into account. Moreover, the particular kind of passive accident that DBQ-violations is associated with, the passive right-of-way violation, has been shown to have a higher rate among young, inexperienced, male drivers (West 1993).

Against the second hypothesis is the finding that there is no clear tendency for reportedly passive accidents of each type to exceed active accidents. The balance of active and passive accidents is very much in line with what would be expected, given unbiased reporting. In addition, the accident categories and the definition of active and passive accident types would make it difficult to distort an accident description to turn it from active to passive. The reporting driver would have to fabricate the accident scenario from scratch.

If the relationship between tendency to commit violations and passive accidents—in particular passive right-of-way violations—is causal, as the above reasoning leads us to believe, it is unclear which aspect of DBQ-violations is responsible. One aspect of DBQ-violations is fast driving speed. Common sense would support the idea that fast driving places individuals at greater risk of colliding with vehicles that are pulling onto or across the carriageway because there is less time to react. However, this hypothesis is contradicted by the failure of DSQ-speed to show any association with passive right-of-way violations. This leaves the possibility that DBQ-violations reflect an approach to driving, not specifically measured by the scale, that carries some risk. One possibility is a wilful disregard of potential hazards, such as a glimpse of a vehicle beginning to emerge from a side road. Another possibility is that DBQ-violations are associated with the tendency not to interpret such situations as potentially hazardous (cf. Brown and Copeman 1975). For example, a driver high on DBQ-violations may fail to appreciate the potential danger of a vehicle edging out and thereby leave himself or herself open to being the "victim" of a passive accident. This is an issue that needs to be explored in further research.

The link between DBQ-violations and active loss-of-control accidents was not unexpected. Common sense suggests that active loss-of-control accidents owe much to driving too fast for the prevailing conditions and this is one component of DBQ-violations. However, this explanation is contradicted by the failure of DSQ-speed to show a clear association with such accidents. There was a tendency for low thoroughness to relate to loss-of-control accidents, but not once age, sex, and driving experience were taken into account. It has already been shown that it is young, male, inexperienced drivers that are at greater risk of active loss-of-control accidents (West 1993). The possibility should now be considered that it is not so much speed per se as failure to adjust speed to the prevailing conditions that is the crucial factor. Individuals high on DBQ-violations may fail to do so because of a wilful lack of concern for their own or others' safety. Alternatively they may neglect to take steps necessary to ensure their safety because they suffer from feelings of invulnerability and the illusion of control demonstrated by McKenna and Lewis (1991). Those low on thoroughness may do so because of a failure to analyse the consequences of failing to adjust their speed. Age and sex may contribute through either or both of the factors just mentioned, and inexperience may contribute by preventing the driver from appreciating the significance of the prevailing road conditions.

The relationship between low thoroughness and active involvement in shunts and right-of-way violations could be explained in terms of a failure of observation. In the case of right-of-way violations, it seems likely that individuals who take a casual approach to decision making would be more likely to fail to take the necessary steps to ensure that there is nothing coming when they pull out of a side road. In the case of active shunts, it should be recalled that approximately half of these involve the vehicle in front having been stationary for some time prior to being hit. A failure to make the careful observations necessary to discriminate a stationary vehicle from the background could well underpin such accidents. With braking shunts, in which the vehicle in front is hit while braking sharply, a failure of observation could similarly contribute, although lack of sufficient headway would also be expected to play a role.

The general failure of fast driving as a characteristic to show clear relationships with specific accident subtypes requires closer scrutiny. As has already been mentioned, it is possible that it is not fast driving per se but failure to adjust speed to the prevailing driving conditions that is important. It is also possible that fast driving is no more than a marker of other characteristics, such as recklessness, that place individuals at increased risk of accidents in general. However, a preliminary study has already shown a relationship between fast driving and what were termed speed-related accidents rather than observation-related accidents (West et al. 1991).

This raises the possibility that the accident classification system adopted in the present study cuts across an important dimension of speed versus observation. Thus some shunts might, for example, be linked to excessive speed on the part of one or both parties, leading to inadequate stopping distances. Other shunts might be observation-related in that the driver of the vehicle behind might have failed to notice that the vehicle in front had stopped. Similarly, most active right-of-way violations would be expected to be observation related in that they tend to involve moving off from a stationary position. However, a certain proportion could be speed related in that a vehicle may be approaching a junction too fast to be able to stop in time. Given the important association between driving speed and accidents in general and the potential value of speed limitation as an accident countermeasure, resolving this issue through further research should be a high priority.

There are two major limitations on any conclusions that may be drawn from these findings. First of all, the samples used are unlikely to be representative of the population of drivers as a whole. In particular, in Study 1, it is clear that the drivers who responded to both accident surveys were at lower risk of accident than those who failed to reply to the second accident survey. This would certainly affect the parameter estimates in the multiple regression analyses, and in general the effect would be to diminish any observed relationships. This is because, with lower accident rates and lower variation in variables associated with accident rates, there would have been some range restriction. Therefore, we believe that where associations were observed between predictor variables and accident rates, these are, if anything, an underestimate of the true relationships in the population. Also, it is difficult to envisage a model in which range restriction would have created the patterns of relationships between behavioural characteristics and different types of accident that were observed. We therefore believe that although the parameter estimates in the multiple regression models are probably incorrect, the pattern of presence or absence of relationships with different types of accident were largely unaffected by the unrepresentativeness of the sample. The same arguments would obviously also apply to Study 2, although in that case there was no follow-up accident survey and therefore less opportunity for biases resulting from nonresponding.

The second potential limitation concerns the extent to which the accidents were correctly classified. Errors could occur as a result of missing or uninterpretable descriptions, misdescriptions, and misclassification by the coders. The last of these would be expected to contribute little noise given that independent coders showed a high level of agreement in the codes assigned (West 1993). It has already been argued that the categories adopted are sufficiently distinct that it would require a complete fabrication of the accident details to move an accident from one category to another. Also, there was little evidence that self-serving biases resulted in disproportionate numbers of accidents being classified as passive. The major obvious source of error came from accidents that could not be classified (West 1993). Limiting analysis only to those for which category assignments could be made may have led to an underestimate of relationships between behavioural characteristics and accident types. Thus, although the system for classifying accidents is crude and no doubt there is a considerable amount of measurement error, there are reasonable grounds for believing that it captures some critical aspects of the accident scenario. In this respect, the measure is not unlike other self-report behavioural measures so widely used in psychology.

The results reported here lead to the conclusion that different behavioural characteristics are associated with risk of different types of accident. They militate against a view that fast driving as a general characteristic has a link with specific accident types as defined in this study. Rather, it seems that disregard for driving norms or some characteristic associated with this may be implicated in losing control of the vehicle and colliding with vehicles that cross one's path, and, of course, the speed at which one is travelling will directly affect the severity of the ensuing crash. Moreover, a casual approach to decision making or some characteristic associated with it may be associated with colliding with vehicles from behind or colliding with a vehicle while pulling onto or across a carriageway. However, the possibility remains that fast driving as a characteristic is directly linked to increased risk of accidents of a type that cuts across the categories adopted in this study. In the light of the present findings it is possible to point to disregard of driving norms and a casual approach to making decisions as potentially useful targets for interventions aimed at reducing the incidence of common accident types.

Further research will be needed to clarify some of the issues raised by the present findings. It will be important to discriminate between general driving speed and propensity to adjust speed to the prevailing driving conditions as possible causes of accidents. It may be difficult to obtain accurate self-report measures of speed adjustment and so in-car observation or simulator studies may be needed. In assessing accident involvement, it will be important to examine the classification system adopted by West et al. (1991) of "speed" versus "observation" accidents and to see how these relate to measures of driving speed and the classification system adopted in the present study. There is also a need to establish what mediates the link between tendency to commit driving violations and increased risk of active loss-of-control accidents and passive right-of-way violations. This could involve relating DBQ-violation scores to behavioural indices that might be expected directly to affect risk of such accidents. For example, it is possible that individuals scoring high on DBQ-violations would make no speed adjustment even having noticed that a vehicle was beginning to pull out in front of them with vision obscured (either through a failure to interpret the situation as hazardous or a lack of concern over potential hazard). Further research is also needed to establish what mediates the link between low decision thoroughness and active shunts and active right-of-way violations. If one assumes that a common characteristic is involved, a tendency not to make sufficiently thorough observations might be a key mediating variable.

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